Introductory Programming with

python

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About the course

Programming

Why should you care, especially in this AI era?

Python

What is it?

Definitions

- **Programming**: way for human to instruct (speak or communicate with) a computer

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- **Algorithm**: set of step-by-step instructions to take some set of inputs and produce a result or output(s)

Algorithm

First step to programming: **wrong algorithm produces wrong results!**

Coding Environment

- Interface for executing computer commands in a given language

Installing the Python Interpreter

To have access to the Python code environment, you need to install the Python interpreter

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We will use the latest Python version: Python 3

```
sudo apt update

sudo apt install python3 python3-pip
```

Check the installation went successfull

```
python3 -V
```

Coding Environment

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- Each environment has its own *vocabulary, syntax, commands, and keywords*

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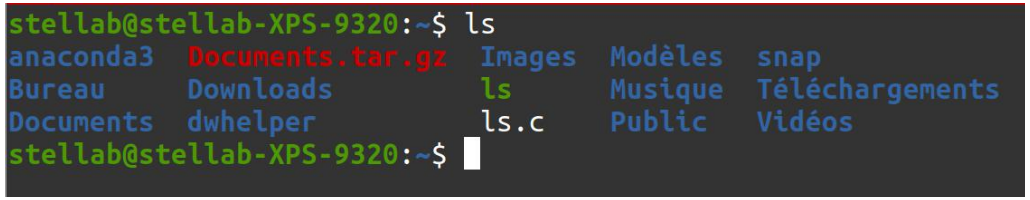
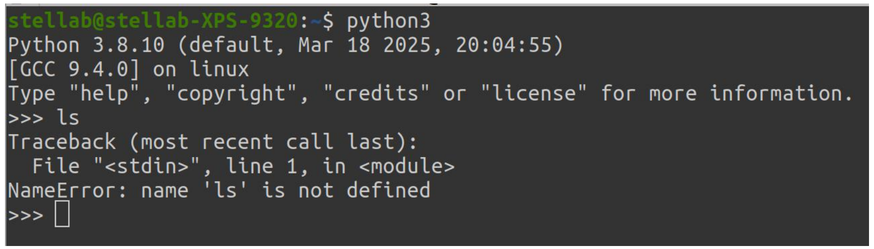
Bash environment



Python environment



Coding Environment

Prompt

Command

Bash environment

```
stellab@stellab-XPS-9320:~$ ls
anaconda3  Documents.tar.gz  Images  Modèles   snap
Bureau     Downloads         ls      Musique   Téléchargements
Documents  dwhelper          ls.c    Public    Vidéos
stellab@stellab-XPS-9320:~$
```

Python environment

```
stellab@stellab-XPS-9320:~$ python3
Python 3.8.10 (default, Mar 18 2025, 20:04:55)
[GCC 9.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> ls
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'ls' is not defined
>>>
```

Code Editor

For simple example and tests, you can easily use the Python environment in your terminal.

To manage complete projects, you will need a more suitable environment, such as code editors: specialized text editors that integrate development tools (e.g., file management, error checking, syntax highlighting, etc.)

Code Editor - VS Code

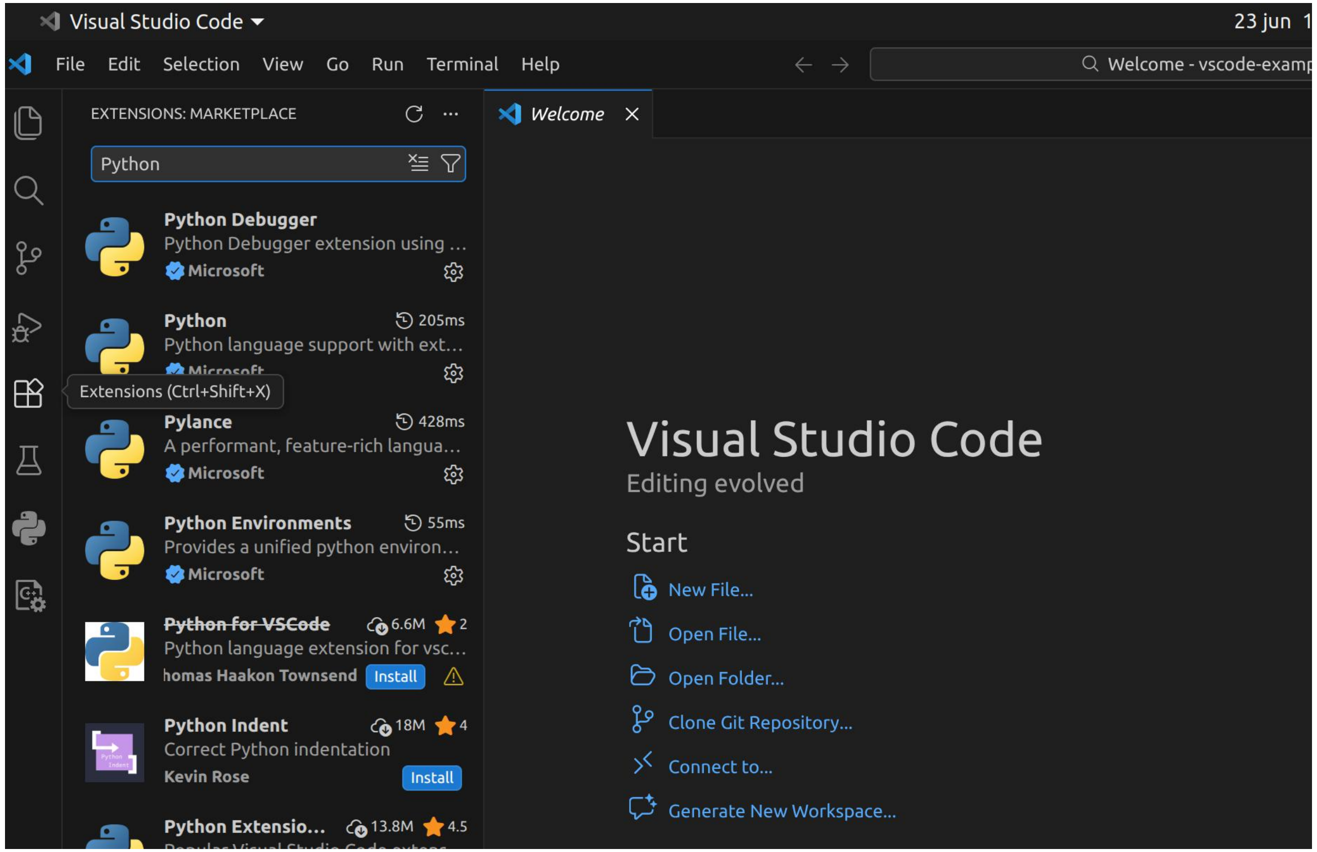
Installation

- Download the installer from the [VSCode Downloads Page]
- This will download a `code_<version>_<architecture>.deb` file that you will install as follows:

```
sudo dpkg -i code_xxx.deb
```

Code Editor - VS Code

Installing the Python extension(s)



Running Python Code in VS Code

Python files use a **.py** extension

Write your first test: create a **hello.py** file and write in it

```
print("Hello, world!")
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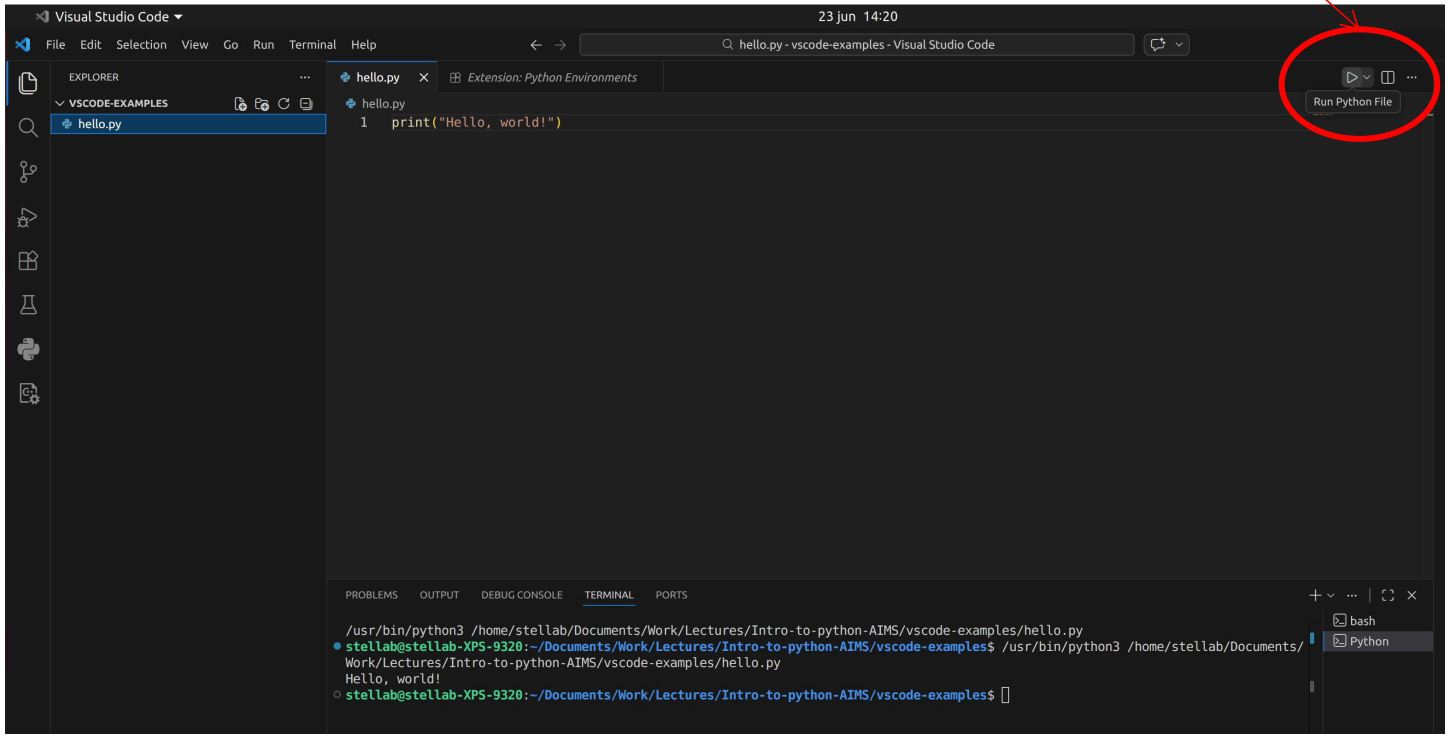
Click on the ‘Run Python File’ icon



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You can also use the vscode terminal (or your built-in terminal):

```
python3 hello.py
```